

Beer's law single point calibration – Practice example 1

A 4.629 g sample of an alloy was dissolved in nitric acid; the resulting solution was transferred to a 100-mL volumetric flask and diluted to volume with water.

A 5.0 mL portion of the solution was pipetted into a 50 mL volumetric flask and diluted to volume with water. This solution was analyzed at 324.7 nm (Cu absorption line) by flame atomic absorption; a net absorbance of 0.119 was observed. A 2.00 mM standard Cu solution yielded a net absorbance of 0.125 on the same AA instrument.

Determine the %Cu in the alloy.

Unk: $\begin{cases} 4.629 \text{ g alloy} \longrightarrow 100 \text{ mL A} \\ 5.0 \text{ mL A} \longrightarrow 50 \text{ mL B} \end{cases}$
 $A_B = 0.119 @ 324.7 \text{ nm}$

Std: $2.00 \text{ mM Cu solution} \longrightarrow A_S = 0.125$

1. use Beer's Law to solve for conc. of Cu in solution B (C_B)
2. use dilution factor and fw of mass alloy to convert to $\frac{w}{w}\%$.

Beer's Law: (k is the slope)

Unk: $0.119 = k \cdot C_B$

Std: $0.125 = k \cdot 2.00$

$\left. \begin{array}{l} 0.119 = k \cdot C_B \\ 0.125 = k \cdot 2.00 \end{array} \right\} \text{ratio} \Rightarrow \frac{0.119}{0.125} = \frac{k \cdot C_B}{k \cdot 2.00}$

$\Rightarrow C_B = \left(\frac{0.119}{0.125} \times 2.00 \right) \text{ mM}$

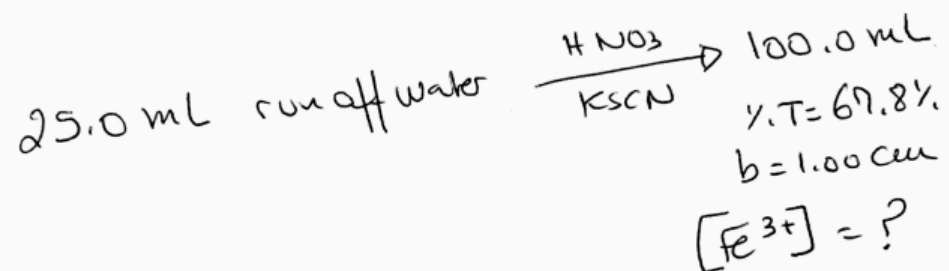
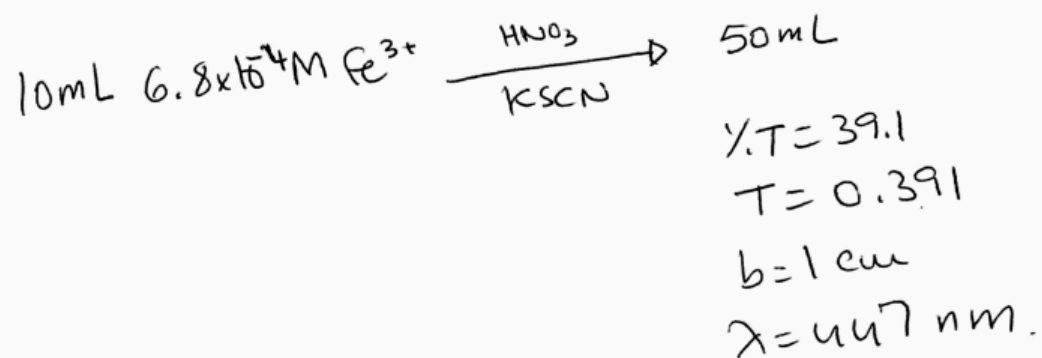
$$\frac{\left(\frac{0.119}{0.125} \times 2.00 \right) \text{ mM Cu}}{\text{mL B}} \times \frac{50 \text{ mL B}}{5 \text{ mL A}} \times \frac{100 \text{ mL A}}{1} \times \frac{1 \text{ mole Cu}}{10^6 \mu\text{mole Cu}} \times \frac{63.55 \text{ g Cu}}{1 \text{ mole Cu}} \times \frac{100}{4.629 \text{ g alloy}}$$

$= 2.61\% \text{ Cu}$

Beer's law single point calibration – Practice example 2

A 10-mL aliquot of a $6.80 \times 10^{-4} \text{ M Fe}^{3+}$ was pipetted into a 50-mL volumetric flask, acidified with nitric acid and treated with excess KSCN to form a red complex with SCN^- ion before diluting to volume with deionized water. The resulting solution transmitted 39.1 % of the light in a 1.00 cm cell at 447 nm, which is a wavelength near the absorption peak for the complex.

A 25.0 mL sample of runoff water was acidified with nitric acid and treated with excess KSCN to form a red complex. The solution was diluted to 100.0 mL in a volumetric flask. The sample was found to transmit 67.8 % of the light in a 1.00 cm cell at the same wavelength. Calculate the concentration of Fe^{3+} in the runoff water.



$$\begin{aligned}
 & \div \left(\begin{aligned} A_s &= -\log T_s = a \times b \times 10 \times \frac{6.8 \times 10^{-4}}{50} \\ A_{rw} &= -\log T_{rw} = a \times b \times \frac{25 \times [\text{Fe}^{3+}]}{100} \end{aligned} \right. \\
 & \frac{-\log 0.678}{-\log 0.391} = \frac{\cancel{a} \times \cancel{b} \times 25 \frac{[\text{Fe}^{3+}]}{100}}{\cancel{a} \times \cancel{b} \times 10 \times \frac{6.8 \times 10^{-4}}{50}} \\
 & [\text{Fe}^{3+}] = 2.25 \times 10^{-4} \text{ M}
 \end{aligned}$$